EXERCISE 11

Invariance of Solution Ratios in Linear ODEs

Original statement · ProblemsCh2.pdf, p. 3

Linear homogeneous • Ratio is constant

PROBLEM STATEMENT

Without solving the ODE, show that if $y_1(t)$ and $y_2(t)$ are any two solutions of

$$\frac{dy}{dt} + p(t)\, y = 0,$$

then the ratio $y_1(t)/y_2(t)$ is constant.

Source: ProblemsCh2.pdf, Exercise 11 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Assume p is continuous on an interval I, and y_1, y_2 are differentiable solutions on I: $y_i' = -p\, y_i$ for $i = 1, 2$. The ratio $r(t) = y_1(t)/y_2(t)$ is defined only where $y_2(t) \neq 0$. So we first have to settle where y_2 can vanish.

Fact needed: a solution of this equation either vanishes identically or has no zeros. This follows from uniqueness (Problem 2.10): if $y_2(t_1) = 0$ at some t_1, then y_2 and the zero function solve the same IVP at t_1, so $y_2 \equiv 0$. We assume $y_2 \not\equiv 0$ (otherwise the ratio is meaningless), hence $y_2(t) \neq 0$ on all of I.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: to show that a function is constant on an interval it is enough to show that its derivative is zero (Mean Value Theorem). The derivative of a quotient can be computed with the quotient rule, and the ODE lets us replace y_1' and y_2' by $-p\,y_1$ and $-p\,y_2$ without solving anything.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

By the formula of the notes with $q = 0$, every solution is $y = y(t_0)\,e^{-P(t)}$ with $P(t) = \int_{t_0}^{t} p$ (uniqueness from Problem 2.10), so $y_1/y_2 = y_1(t_0)/y_2(t_0)$ wherever $y_2(t_0) \neq 0$. The problem asks for the same conclusion without solving, which is the quotient-rule argument below.

PHASE 3: STEP-BY-STEP DERIVATION

Apply the quotient rule to $r = y_1/y_2$, on I where $y_2 \neq 0$:

$$r'(t) = \frac{y_1'(t)\,y_2(t) - y_1(t)\,y_2'(t)}{y_2(t)^2}.$$

Substitute $y_1' = -p\,y_1$ and $y_2' = -p\,y_2$:

$$r'(t) = \frac{(-p\,y_1)\,y_2 - y_1\,(-p\,y_2)}{y_2^2} = \frac{-p\,y_1y_2 + p\,y_1y_2}{y_2^2} = 0.$$

So $r' \equiv 0$ on the interval I, and by the Mean Value Theorem r is constant: $y_1(t) = c\,y_2(t)$ with

$$c = \frac{y_1(t_0)}{y_2(t_0)} \quad \text{for any } t_0 \in I.$$

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Consistency with the explicit solution (Problem 2.10): $y_i = y_i(t_0)\,e^{-P(t)}$ with $P(t) = \int_{t_0}^{t} p\,ds$, so $y_1/y_2 = y_1(t_0)/y_2(t_0)$, independent of t. Example: $p = 2t$, $y_1 = 3e^{-t^2}$, $y_2 = e^{-t^2}$, ratio 3.

Interpretation: the solution set of $y' + py = 0$ is a one-dimensional linear space, spanned by e^{-P}. Any two non-trivial solutions are proportional.

Limits: the result requires $y_2 \neq 0$ on the interval, which is guaranteed by the Fact in Phase 1. Without linearity (for example $y' = y^2$) the argument breaks down: the term $-p\,y_i$ is replaced by a non-proportional expression and r' does not vanish.

RESULT

$\frac{d}{dt}\left(\frac{y_1}{y_2}\right) = 0$, hence $y_1/y_2 \equiv c$ is constant.